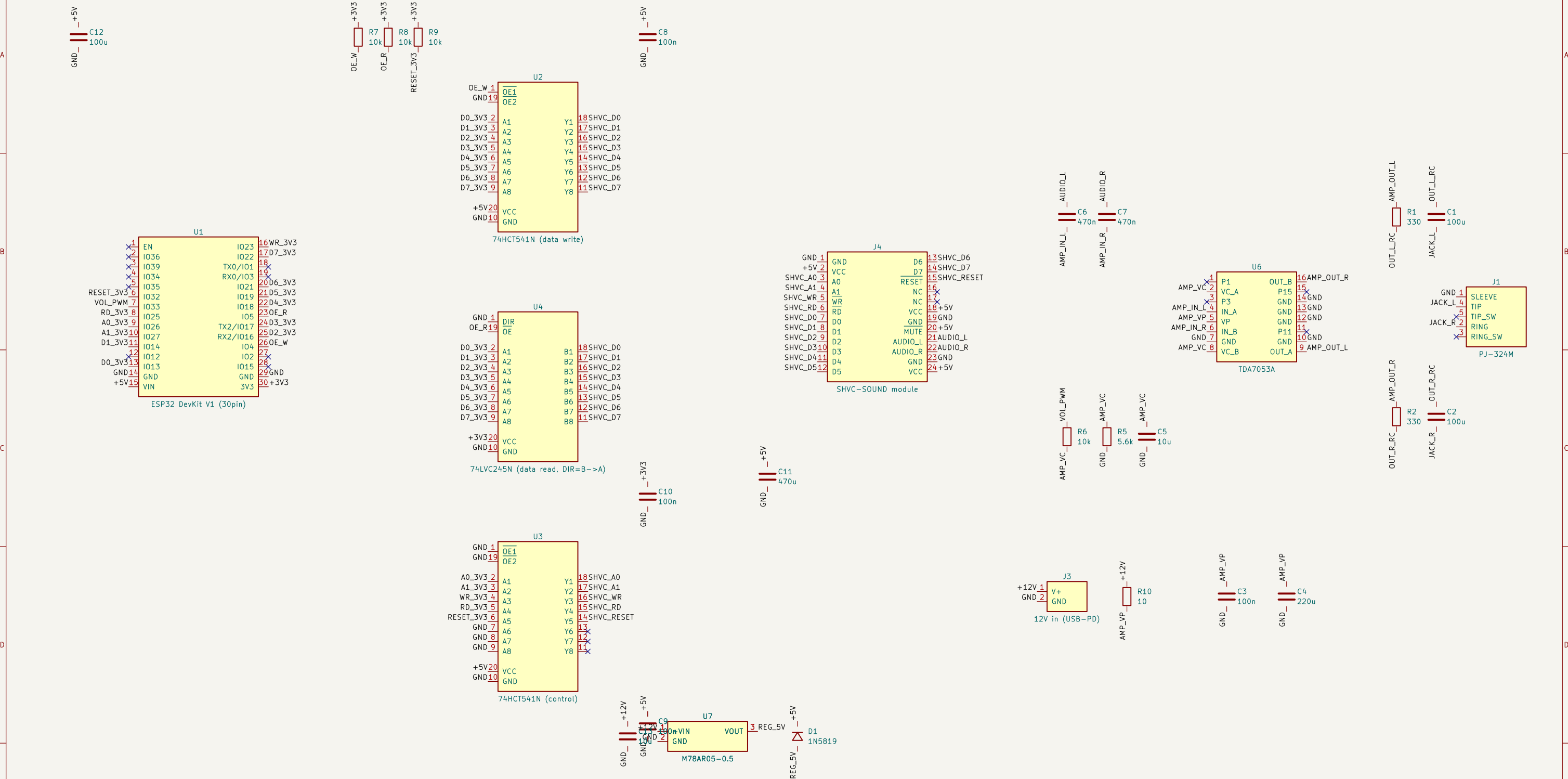


SHVC-SOUND player – ESP32 version (74HCT541N x2 + 74LVC245N + TDA7053A)



Data bus: U2(HCT541, write 3.3V->5V) and U4(LVC245 DIR=GND, read 5V->3.3V) are in parallel.

Firmware must never assert OE_W and OE_R (both active low) at the same time.

R7/R8 keep both buffers disabled while the ESP32 boots. R9 keeps /RESET inactive.

ESP32 DevKit V1 30pin (DOIT type). GPIO16/17 are PSRAM on WROVER boards – use WROOM-32.

Amp circuit copied from SHVC-SOUND-KiCad (Arduino version). R5 3.3k -> 5.6k for 3.3V PWM, ferrite bead FB1 -> R10 10 ohm.

TDA7053A pin names are inferred from the existing schematic; check the datasheet.

12V (USB-PD) -> U7 M78AR05-0.5 -> D1 1N5819 -> +5V. USB on ESP32 can stay connected (board has VIN diode – check yours).

Tie all grounds together at one point. Use electrolytics with correct polarity for C1/C2/C4/C5/C11/C12.